

SHIVANSH SINGH

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EDUCATION

University of Toronto | BAsC - Electrical & Computer Engineering

Graduated April 2026

- **Minors:** Artificial Intelligence Engineering | Robotics & Mechatronics
- **Relevant Coursework:** Algorithms & Data Structures, Software Design and Communication

TECHNICAL SKILLS

Programming: Modern C++ (17/20/23), JavaScript/TypeScript (React, Node.js), Python, SQL (PostgreSQL), Go, Java

Systems & Architecture: Distributed Systems, Lock-Free Concurrency, Full-Stack Development, Microservices, API Design (REST, gRPC), Real-Time / Embedded

Tools & Infrastructure: Docker, Git, CI/CD, AWS (EC2, S3), Prometheus, Grafana, Linux, ROS2, Valgrind, GDB, VTune, OpenVINO, PyTorch, CUDA, AVX-512 SIMD

WORK EXPERIENCE

Software Engineering Intern | Ribbit (Autonomous Aircraft)

Sep. 2024 – Aug. 2025

- Engineered real-time C++17 service layers for autonomous flight vehicles using zero-copy data pathing and lock-free ring buffers, sustaining sub-100ms deterministic message-processing cycles.
- Wrote embedded drivers for safety-critical sensors/actuators with template metaprogramming for zero-overhead hardware abstraction, and built multi-threaded RF audio streaming with custom buffer management that eliminated jitter over low-bandwidth radio.
- Built system-level telemetry monitoring (Prometheus, Grafana) for real-time incident response and end-to-end observability across multi-node distributed flight systems.

Software Lead - Deep Learning Acceleration | aUToronto (UofT Self-Driving Car Team)

Sep. 2023 – April 2024

- Refactored latency-critical perception pipelines from Python to Modern C++ ROS2 nodes within a large-scale codebase, eliminating GIL constraints and increasing throughput by 140%.
- Profiled with Intel VTune and Valgrind to fix cache misses and heap bottlenecks, and containerized CV pipelines with Docker + OpenVINO quantization, boosting team testing velocity by 50%.

Software Lead - UAV Multicopter | UofT Aerospace Team (UTAT)

Sep. 2025 – Present

- Architected a resilient, event-driven IPC framework for autonomous flight nodes, ensuring fault-tolerant crash resilience, asynchronous state synchronization, and high uptime across mission-critical cycles.
- Defined the test strategy and integration suite for flight-node protocols, improving system reliability and reducing field-failure rates across high-bandwidth autonomous workflows.

PROJECTS

Velox-LOB: Sub-Microsecond Matching Engine | C++20, Lock-Free, AVX-512 SIMD

Jan. 2026 – Present

- Built a limit order book sustaining 1M+ order updates/sec using Data-Oriented Design for L1 cache-hit maximization and a custom fixed-size block-pool allocator that eliminated heap-fragmentation latency spikes.
- Engineered a lock-free single-producer/single-consumer queue via `std::atomic` with AVX-512 SIMD for parallel price-level updates, achieving a 4x reduction in matching latency and sub-microsecond processing.

Autonomous Robotic Service Dog (Capstone, Lead Architect) | ROS2, C++, OpenCV, Inverse Kinematics

Sep. 2025 – Present

- Led a distributed ROS2 stack for an 8-DOF quadruped integrating perception, planning, and control, with custom QoS profiles for deterministic multi-node communication.
- Built a closed-loop visual-servoing pipeline (OpenCV state estimates → velocity commands) and an IK solver for task-space leg actuation, enabling adaptive real-time human tracking.

Rescue Ranger [Devpost] | React, Node.js, REST, AWS (EC2, S3), Arduino, C++

Jan. 2023

- Built a first-place autonomous rover (1st of 150+ teams, UofTHacks X) with an Arduino chassis, CV-based survivor identification, and real-time obstacle avoidance.
- Designed a REST API backend on AWS EC2 to stream live rover telemetry; stored mission data and images in S3, powering a React dashboard with real-time Google Maps visualization.

EXTRACURRICULARS

Hackathons: UofTHacks X 1st Place, UTEK 2nd Place, HackTheNorth 2023 [[Devpost](#)], NSBEHacks [[Devpost](#)]

Varsity Athlete: UofT Wrestling & Rowing; MMA/BJJ bronze medallist (Canadian nationals); former national-level swimmer (India).
15+ hrs/week competitive training alongside full-time engineering.